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Application of the Deep Learning Integrated Framework CEEMDAN-GRU-Informer in Financial Time Series Prediction

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Abstract

In the realm of financial markets, accurately predicting stock prices presents both a significant research challenge and a practical necessity for investors and financial institutions. While traditional methods based on economic and financial theories provide a foundational framework, their limitations in capturing non-linear and dynamic patterns necessitate the exploration of advanced predictive techniques. This study addresses the challenges associated with financial time series prediction by proposing the CEEMDAN-GRU-Informer model within a deep learning framework. By incorporating the latest Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) technology, the original financial time series is decomposed to extract price variation information across different time scales. The complexity of each decomposed sequence is assessed using the Sample Entropy (SE) method, facilitating the selection of an appropriate predictive model. In the model prediction phase, a hybrid approach combining Gated Recurrent Unit (GRU) and Informer architectures is employed to enhance forecasting accuracy, leveraging various data-driven techniques to adapt to the complexity of the sequences. This research aims to significantly improve forecasting capabilities, offering stakeholders valuable insights for decision-making in volatile and dynamic financial environments. Empirical studies demonstrate that, within the SZ1 and SZ6 datasets, the root mean square error (RMSE) of the proposed model is 0.0007 and 0.0009, respectively, indicating substantial improvements over other hybrid models and highlighting its suitability for financial time series forecasting.

Keywords Stock price prediction · CEEMDAN · Sample entropy · Hybrid model · Informer

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1 Introduction

In the financial market, accurately predicting the future prices or returns of target assets is essential for minimizing decision-making risks and effectively forecasting asset trends. Consequently, stock prediction has garnered significant interest due to its critical role in facilitating informed decisions and reducing uncertainties in asset management (Yilmaz & Yildiztepe, 2024; Xu et al., 2021). Traditional analytical approaches, grounded in economics and finance, primarily employ fundamental and technical analysis to forecast fluctuations in stock prices (Caraiani, 2020; Chen et al., 2023).

Fundamental analysis focuses on evaluating the intrinsic value of stocks and examining market factors that influence them, including exchange rates, industry regulations, inflation, and international relations (Nti et al., 2020; Ko & Chang, 2021). In contrast, technical analysis emphasizes the study of stock price trends, trading volumes, and investor psychological expectations to assess the trajectory of individual stocks or overall market indices. Despite the widespread use of these methods among various institutions and individual investors, they exhibit certain inaccuracies (Harel & Harpaz, 2021; Yu & Huang, 2021). These inaccuracies can be attributed to the combined effects of various factors, such as the global landscape, domestic and international economic conditions, and stock market operations. Notably, the accuracy of such predictions is heavily reliant on the professional competence of analysts, which makes achieving precision challenging (Vanaga & Biruta, 2020; Lin & Ha-Kyun, 2020).

With the rapid advancement of machine learning technologies and the inherent challenges of modeling in economics, data-driven methods have emerged as effective tools for predicting stock price fluctuations (Ma et al., 2022). Traditional statistical models rely on statistical inference to describe and assess relationships between variables. However, these models are based on the assumption of linear relationships within their structure, which limits their ability to capture the nonlinear characteristics inherent in financial data (Haan et al., 2016; Bustos & Pomares, 2020). Given that financial markets are highly noisy, nonparametric dynamic systems, their time series often exhibit traits such as high noise levels, non-stationarity, and dynamic behavior (Chen et al., 2021a, b; Staffini, 2022).

In response to these challenges, researchers have proposed deep learning-based prediction methods that can more effectively capture price changes. For instance, Lu et al. (2020) utilized Convolutional Neural Networks (CNNs) for feature extraction from input data, achieving an improvement in prediction accuracy by 3% to 20% through the application of features extracted by Long Short-Term Memory (LSTM) networks. Similarly, Chen et al., (2021a, b) constructed feature vectors as input for Bidirectional Long Short-Term Memory (BiLSTM) models, incorporating the Efficient Channel Attention (ECA) mechanism to enhance feature learning. Additionally, Ding & Qin (2020) developed an LSTM deep recursive neural network model capable of managing multiple inputs and predicting various stock prices with an accuracy exceeding 95%.

Despite these advancements, many of these approaches often employ individual models for price prediction, overlooking the importance of considering long-term trends, periodic fluctuations, and local noise within price movements. This narrow focus can limit their effectiveness in addressing the complexities present in diverse financial time series.

To address the limitations of using a single model for complex data prediction, researchers have explored the implementation of hybrid models that integrate multiple independent models to enhance predictive accuracy (Ji et al., 2021; Wang et al., 2021). For instance, Lv et al. (2022) proposed a hybrid approach that combines attention mechanisms with Gated Recurrent Units (GRU). This method assesses the influence of various time-price information on the final prediction value through attention scores, resulting in a more effective model structure and improved predictive performance. Similarly, Niu et al. (2020) demonstrated that combining Variational Mode Decomposition (VMD) with Long Short-Term Memory (LSTM) significantly reduced the discrepancy between actual and predicted values. Wang & Wang (2017) developed a prediction model that integrates Empirical Mode Decomposition (EMD) with a Stochastic Time-Intensity Neural Network. This model decomposes the original time series into various subsequences and performs targeted training based on the frequency characteristics of different intrinsic mode functions. In another innovative approach, Hossain et al. (2021) utilized the EMD decomposition method to propose a novel fusion technique that combines Autoregressive Integrated Moving Average (ARIMA) and Exponential Weighted Moving Average (EWMA). In this model, the prediction method is chosen based on the frequency of the decomposed sequence, thereby accounting for the complexity of the data. Collectively, these approaches highlight the effectiveness of hybrid models in enhancing predictive accuracy in financial time series forecasting.

While hybrid models have shown improvements in effectiveness over individual models, they still face certain limitations. The Variational Mode Decomposition (VMD) method, a prominent decomposition technique, encounters challenges in selecting the appropriate number of modes during decomposition. This choice significantly affects the decomposition results, and currently, no universal method exists to determine the optimal modal number. In contrast, Empirical Mode Decomposition (EMD), as an adaptive decomposition approach, can automatically adjust the decomposition process based on the local characteristics of the signal, thus addressing some limitations of VMD. However, EMD methods are often hindered by issues such as pattern overlap (Lv et al., 2022; Tian et al., 2023). The latest advancement, Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), incorporates integrated thinking, leading to enhancements over EMD. Nevertheless, these improvements have not been swiftly applied to financial sequences.

At the same time, the attention and dual-head mechanisms commonly employed in existing methods still rely on recurrent neural networks, which struggle with gradient vanishing and exploding issues. This poses limitations in processing high-complexity data and managing long-term dependencies. In contrast, neural networks based on the Transformer's self-attention mechanism have made significant strides in forecasting complexity and long-term dependencies compared to traditional methods (Vaswani et al., 2017). Building on this, Zhou et al. (2021) introduced the Informer model, which utilizes a sparse self-attention mechanism and demonstrates exceptional performance in capturing global information. This model is particularly well-suited for stock price prediction, effectively handling high-complexity data extracted through hybrid methods (Wang et al., 2023).

In summary, this paper presents a novel deep learning theoretical framework for forecasting financial time series and constructs a CEEMDAN-GRU-Informer model based on this framework. The GRU is adept at handling simpler or computationally efficient sequence components, while the Informer excels in managing complex sequence components that require capturing long-term dependencies. By integrating the preprocessing advantages of CEEMDAN, the entire framework synergizes the strengths of all three components, enhancing its flexibility and efficiency in addressing various prediction tasks. Additionally, sample entropy (SE) is introduced to establish a threshold for assessing high and low complexity. The prediction accuracy of individual decomposed sequences is further improved by employing multiple data-driven methods, tailored to accommodate differences in sequence complexity, significantly enhancing the prediction accuracy of stock indices. The innovations of this paper are as follows:

1) A new learning paradigm, CEEMDAN-GRU-Informer, is proposed for predicting stock prices one day in advance. The preprocessing of stock price data utilizes CEEMDAN and SE to reduce the complexity of the original data.
2) In the model prediction section, GRU and Informer are combined for predicting decomposed sequences. By analyzing the sample entropy of the decomposed sequences and the training outcomes, a threshold is determined to differentiate between high and low complexity. This allows for the selection of different prediction methods based on complexity, improving overall prediction accuracy by leveraging the adaptability of deep learning models to varying sequence complexities.
3) This study innovatively integrates CEEMDAN, Informer, and GRU into a unified framework for stock price forecasting. This interdisciplinary technological advancement aims to harness the strengths of each model, enhancing the ability to capture complex market dynamics, particularly when dealing with nonlinear, non-smooth, and highly noisy financial time series data.

The structure of the paper is as follows: Section "Methodology" outlines the theoretical foundation adopted in this study. Section "Data Processing" details the data selection and processing principles. Section "Results and Discussion" provides an in-depth analysis of the predictive results obtained through the proposed method, comparing them across various datasets with results from multiple methodologies. Finally, Section "Conclusion" presents the corresponding conclusions.

2 Methodology

2.1 The Hybrid Forecasting Model

Figure 1 illustrates the overall workflow of the proposed method.

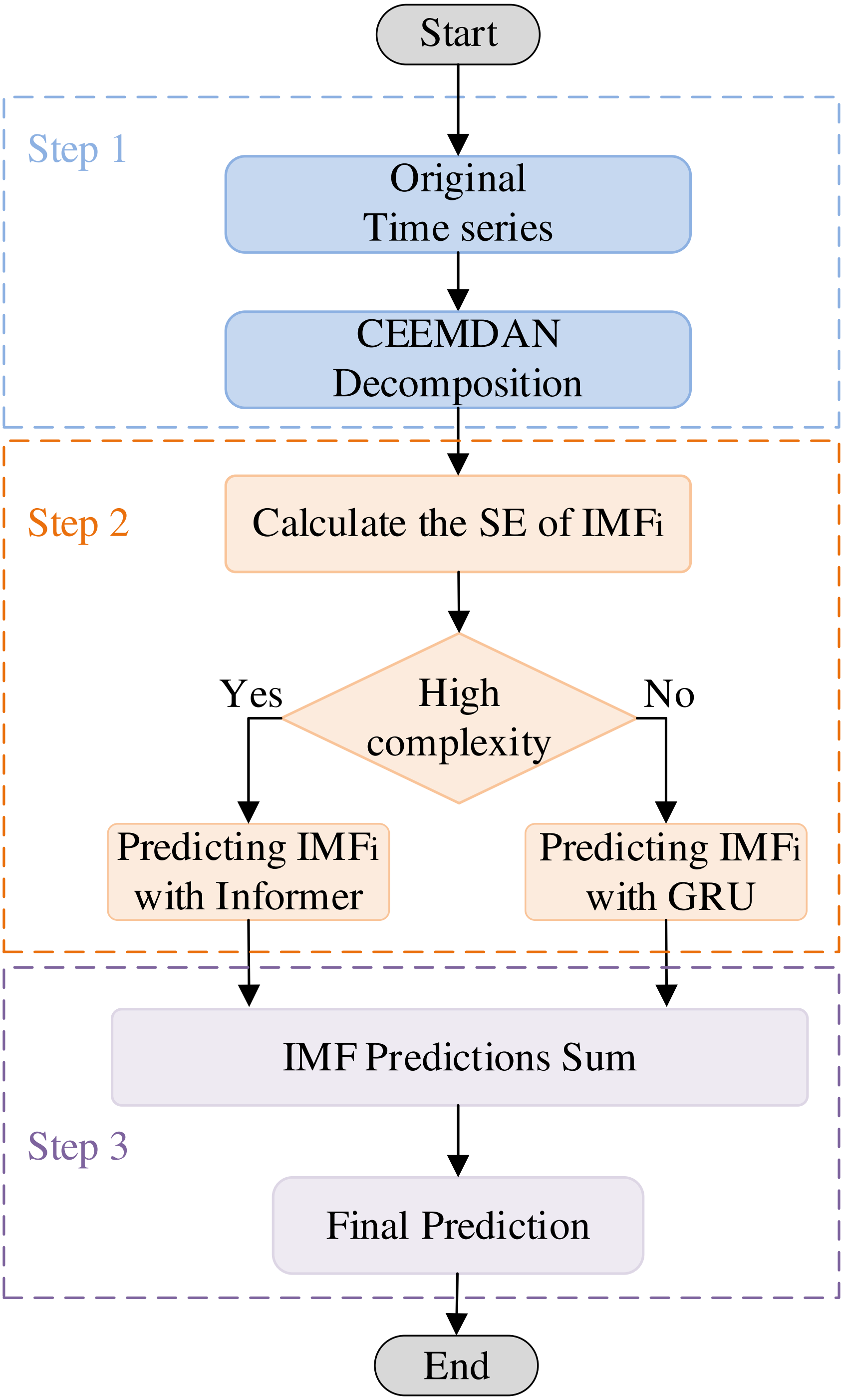


Fig. 1 Framework diagram of prediction method

This study begins by employing CEEMDAN to decompose financial time series data into multiple Intrinsic Mode Function (IMF) sequences. Each IMF sequence captures unique patterns of price variation across different time scales. The complexity of each IMF sequence is assessed using the Sample Entropy (SE) method, where higher SE scores signify greater complexity. Based on these evaluations, a tailored prediction approach is adopted: the Gated Recurrent Unit (GRU) model is utilized for sequences with lower complexity, while the Informer model is applied to those with higher complexity. Predictions generated by both the GRU and Informer models for all IMF sequences are then aggregated to produce the final prediction. This methodology aims to leverage the strengths of each model according to the complexity of the sequences, thereby enhancing overall predictive accuracy in financial time series forecasting.

2.2 CEEMDAN Decomposition

The Adaptive Noise-Embedded Empirical Mode Decomposition (CEEMDAN), as an improved algorithm of Empirical Mode Decomposition (EMD), has been successfully applied to signal denoising. In contrast to wavelet and Fourier decomposition algorithms, the CEEMDAN algorithm does not require assumptions of data stationarity and linearity, thus offering a broader range of applications and clear advantages in handling non-stationary time-series data.

2.3 Sample Entropy Analysis

Sample entropy, proposed by Pincus, serves as an enhancement over traditional entropy measures. Although traditional approximate entropy is straightforward, it can be affected by noise when applied to large datasets. In contrast, sample entropy offers a more robust measure of complexity by addressing some of the limitations associated with approximate entropy. This measure assesses the degree of irregularity in a time series, specifically the frequency of complex patterns occurring within the data. A lower sample entropy indicates a more regular pattern, whereas a higher sample entropy signifies greater irregularity in the variation of the series.

In this paper, we calculate the sample entropy for each Intrinsic Mode Function (IMF) during the training phase, utilizing the decomposition results from the training set. We then identify the optimal outcomes by selecting the Informer and GRU models based on the training set evaluation results, tailored to the different IMFs. A threshold is established using the sample entropy corresponding to the bounded IMFs. This threshold is subsequently applied in the testing phase to differentiate between high and low complexity: values above this threshold are classified as high complexity, while those below it are categorized as low complexity.

2.4 GRU

GRU, a variant of Recurrent Neural Networks (RNNs), employs a gating mechanism akin to LSTM networks to manage the updating and retention of hidden states. However, GRU simplifies this structure by incorporating only two gates: the reset gate and the update gate. Consequently, GRU offers faster prediction speeds and robust prediction accuracy compared to LSTM. For data with more regular patterns or relatively simple sequences, GRU can capture these features well and make accurate predictions, so this paper adopts GRU to predict low-complexity IMF (Fig. 2).

2.5 Informer

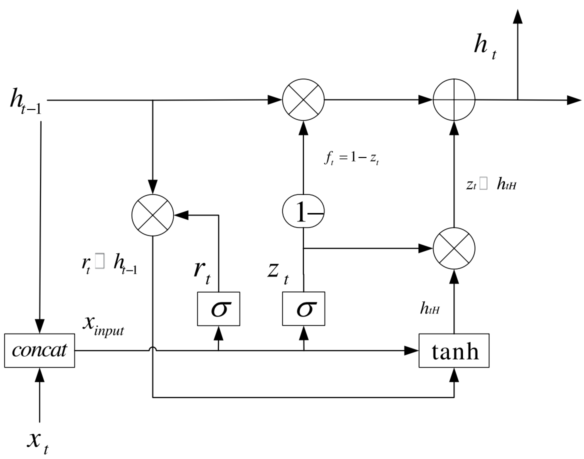
Informer is a Transformer model specifically designed for Long Series Time Series Forecasting (LSTF). Compared to traditional Transformers, Informer offers three unique features. First, it utilizes the ProbSparse self-attention mechanism, which ensures low time complexity and reduced memory usage. This allows it to effectively capture long-term dependencies within the sequence. Second, through self-attention distillation techniques, Informer efficiently handles extremely long input sequences. Finally, its generative decoder can predict the entire long time series in a single step, significantly enhancing efficiency during the forecasting process. Extensive experiments on large-scale datasets have demonstrated that Informer excels in addressing LSTF challenges, providing an efficient and accurate solution that overcomes the limitations of traditional Transformer models (Zhou et al., 2021). The specific architecture of Informer is illustrated in Fig. 3.

3 Data Processing

3.1 Data Selection

To minimize the potential impact of financial market maturity on the predictive performance of deep learning models, this article selects three distinct categories of representative stock indices: the S&P 500 Index (SPX.GI), the Dow Jones Industrial Average (DJI.GI), the Shenzhen Component Index (399001.SZ), the Shanghai Composite Index (000001.SH), the Hang Seng Index (HSI.HI), and the Growth Enterprise Market Index (399006.SZ).

Fig. 2 GRU structure diagram



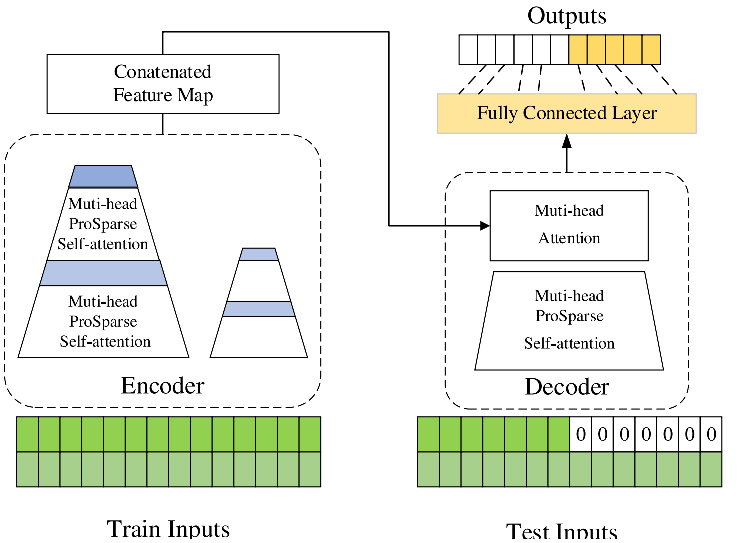


Fig. 3 Informer working principle

As a hallmark of advanced financial markets, the New York Stock Exchange boasts comprehensive market infrastructure, mature operating systems, and extensive experience. Specifically, the S&P 500 Index and the Dow Jones Industrial Average are recognized as the most representative indices in this market, both classified as belonging to developed markets. In contrast, the financial markets in mainland China remain in a developmental phase, thus categorizing them as emerging markets (Ma & Yan, 2022). This study selects the Shenzhen Component Index, the Shanghai Composite Index, and the ChiNext Index to represent these emerging markets. Additionally, the Hong Kong stock market occupies a developmental level between advanced and emerging markets, leading to the selection of the Hang Seng Index as a representative of relatively developed markets.

Based on these considerations, this research employs these six indices to provide a comprehensive characterization of international markets, aiming to investigate the robustness of deep learning prediction models across different stages of financial market development.

3.2 Data Preprocessing

The daily closing price data for the six stock indices from August 30, 2011, to April 27, 2018, has been obtained, as illustrated in Fig. 4. To validate the proposed effectiveness, the selected data is processed to derive the logarithmic return rate rt for the t-th day, represented as follows:

$$\mathrm{r_t} = \ln P_t - \ln P_{t-1} \tag{1}$$

where P_t is the closing price on day t.

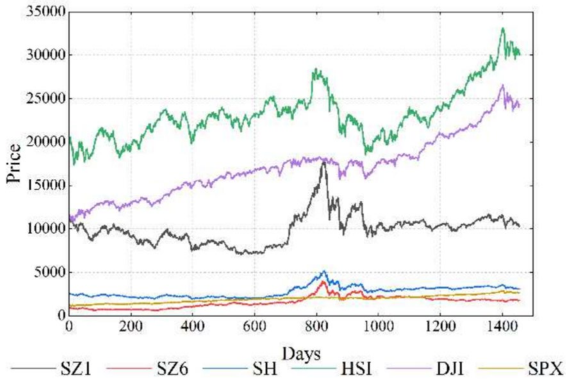


Fig. 4 Original stock price

Table 1 Descriptive statistical results of data

	SZ1	SZ6	SH	HSI	DJI	SPX
Mean	−6.64E-05	0.000438	0.000123	0.00029	0.000512	0.000544
Max	0.07224	0.075826	0.063693	0.06987	0.045968	0.055124
Min	−0.10319	−0.10949	−0.08873	−0.06162	−0.04714	−0.04184
Std-dev	0.017643	0.021734	0.014955	0.011936	0.0088	0.009139
Var	0.000311	0.000472	0.000224	0.000142	0.0000774	0.0000835
Skewness	−0.67828	−0.51256	−0.90226	−0.30783	−0.25188	−0.125
Kurtosis	4.17256	2.356894	6.643692	3.409549	3.68181	3.888445
C-V	−265.696	49.60274	121.4773	41.22695	17.19446	16.81514
JB	1173.094	402.9949	2886.399	731.9441	841.9174	925.5539
ADF	−12.1319	−11.8108	−7.83657	−38.7765	−17.6063	−18.0349

3.3 Data Descriptive Statistics

To thoroughly analyze the statistical properties of the utilized data and assess its usability, we calculated the relevant statistics for the selected datasets, as presented in Table 1 below.

The analysis reveals that the average daily market returns are predominantly close to zero. Specifically, the SPX and DJI indices exhibit the highest average daily returns at 0.0543519% and 0.0511797%, respectively, indicating their strong growth potential. In contrast, the SZ1 index shows a slight negative return of approximately −0.0006%. This suggests that, from a long-term perspective, the SPX and DJI indices demonstrate the most favorable growth trends, while the SZ1 index appears somewhat subdued.

Regarding volatility, the HSI, DJI, and SPX indices exhibit relatively low volatility, with the DJI showing the highest stability, evidenced by its smallest standard deviation of 0.00880007. Conversely, the SZ1 and SZ6 indices are more volatile, particularly the SZ1, which has a standard deviation of 0.0176433, reflecting more pronounced day-to-day fluctuations in these market yields.

Skewness analysis indicates that the distribution of all market returns is left-skewed, suggesting that extreme downward movements occur more frequently than upward movements. Furthermore, the kurtosis values are generally high, especially for the Shanghai Composite Index (SH), indicating that the yield distributions exhibit sharp peaks and thick tails, which implies a higher probability of extreme values. Additionally, the Jarque–Bera (JB) test significantly rejects the normal distribution assumption, while the Augmented Dickey-Fuller (ADF) test confirms that the yield series of these markets are stationary, making them suitable for time series analysis.

The dataset is divided into training and testing sets in an 80:20 ratio. During the training phase, CEEMDAN decomposition is applied to the entire training dataset. In the testing phase, sliding decomposition is conducted using a sliding window approach, where the window size matches the length of the training set data. This sliding window is re-decomposed for each prediction step, and the final prediction results are utilized to evaluate the model's performance.

4 Results and Discussion

This section establishes various evaluation metrics to assess the predictive performance of the proposed method. Using the Shenzhen Component Index (399,001.SZ, abbreviated as SZ1) as a case study, we analyze the performance of the CEEMDAN decomposition, sequence sample entropy, and the predictive outcomes of the proposed fusion method. Ultimately, we validate the superiority of the proposed fusion method across multiple datasets through various analytical approaches. Evaluation metrics.

The forecast errors between predicted values and real values are quantified based on Mean Absolute Error (MAE), Root Mean Square Error (RMSE). The calculation principles for each metric are as follows:

$$MAE = \frac{1}{N}\sum_{t=1}^{N}|r_t - p_t| \tag{2}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{t=1}^{N}(r_t - p_t)^2} \tag{3}$$

where N means the total number of test data, r_t represents the real logarithmic return for the t-th day in the test data, while p_t represents the predicted logarithmic return for the test data.

4.1 Model Confidence Set Test

To further examine whether there is a statistically significant difference between the model predictions, this paper uses the model confidence set (MCS) method in combination with several of the above evaluation metrics to test the statistical accuracy of the model predictions. According to the basic principle of MCS test, if the P-value of the model MCS test is less than 0.1, it means that the model is a model with poor predictive ability, while the model with P-value more than 0.1 is a model with good predictive ability and will survive the MCS test.

Scholars have studied the commonly used $d = 2$ and $B = 10000$ (d is block length, B is the number of simulations) as a control parameter, choose $\alpha = 0.1$ significance level (MCS p-value is greater than 0.1 that is, better predictive ability of the model, only better predictive ability of the model can be through the MCS test, retained until the other models are all eliminated), p-value the higher the then it indicates the higher prediction accuracy of the model.

4.2 CEEMDAN Decomposition Results

Direct forecasting poses a significant challenge due to the inherent volatility of stock price series. Therefore, this study employs the CEEMDAN algorithm to partition the original series into stable intrinsic modal components. Figure 5 depicts the CEEMDAN decomposition results for the SZ1 dataset, revealing 11 intrinsic modes. From Fig. 5, it can be seen that IMF0-IMF2 exhibit wide numerical ranges but volatile trends, likely reflecting the impact of short-term market factors. On the other hand, IMF3-IMF10 show broader numerical ranges with clear non-linear trends, suggesting influences from long-term or cyclical market forces. IMF10 demonstrates a stable downward trend but has a narrower numerical range, providing limited information on overall price changes compared to itstrend impression.Although the exact correspondence between the decomposed IMF series and the diversified influencing factors of financial time series is not yet clear, it can still be considered that the trend of these IMF sequences can provide some explanation for the influencing factors in

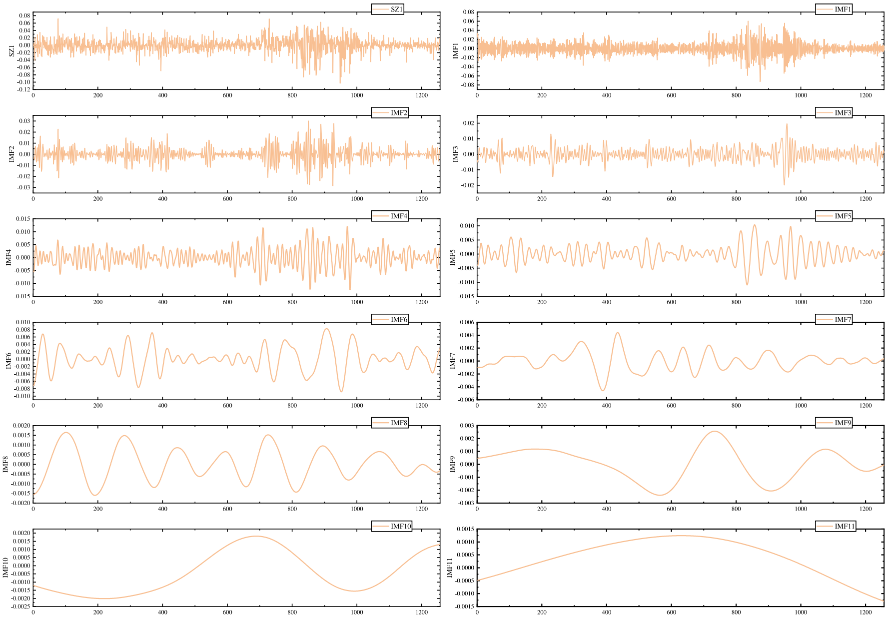


Fig. 5 IMF decomposition results of SZ1

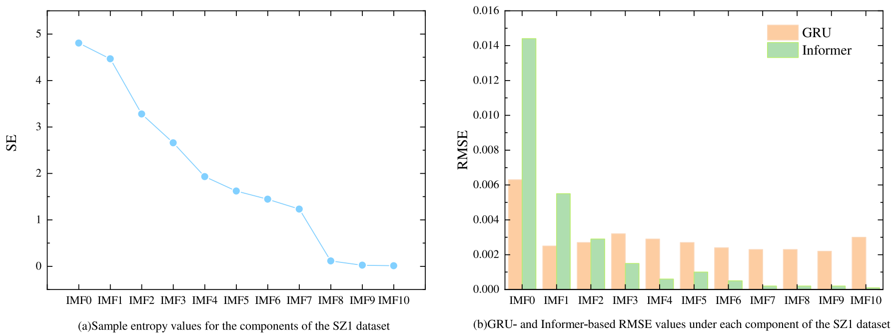


Fig. 6 SE values for each component and RMSE for different prediction methods

Table 2 Sample entropy of IMF sequence of SZ1

Column	IMF0	IMF1	IMF2	IMF3	IMF4	IMF5	IMF6	IMF7	IMF8	IMF9	IMF10
SE	4.8047	4.4645	3.2788	2.6579	1.9302	1.6178	1.4427	1.2301	0.1136	0.0235	0.0097

the market, demonstrating the superiority of the CCEMDAN decomposition method used in this study.

4.3 Sample Entropy Results

The components after CEEMDAN decomposition tend to have different statistical properties. Among them, high-frequency components usually contain more short-term fluctuations and noise, while low-frequency components are more reflective of long-term trends and patterns. Therefore, in order to effectively classify the categories of components, this paper adopts SE to measure the complexity of components.The sample entropy results of SZ1 are shown in Fig. 6 and Table 2. Figure 6 shows:the sample entropy of the decomposed sequences from IMF0 to IMF10exhibits an overall decreasing trend, indicating a gradual reduction in the complexity of each IMF sequence.

Based on the SE characteristics of each sequence decomposed by SZ1 data and the prediction errors of GRU and Informer, the threshold value in this paper is chosen as 3.2788, and Informer is used for prediction when the sample entropy of the sequences is greater than 3.2788, and GRU is used for prediction when the sample entropy of the sequences is less than 3.2788 in the testing stage. The total prediction results obtained are then presented in Fig. 7.

4.4 Proposed Method Prediction Result

Research has demonstrated that selecting a model tailored to the characteristics of specific frequency components can enhance prediction accuracy. The GRU model excels in handling long-term dependencies in time series data, while Informer is particularly effective for high-frequency and complex time series. This paper employs a combination of GRU and Informer to predict low and high frequencies, respectively,

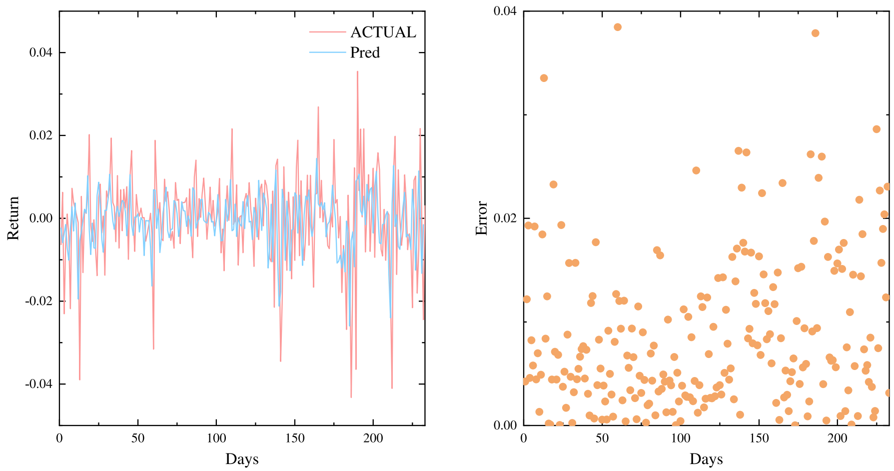


Fig. 7 Final prediction result of SZ1 by the proposed method

Table 3 Prediction RMSE error of IMF sequence of SZ1

	IMF0	IMF1	IMF2	IMF3	IMF4	IMF5	IMF6	IMF7	IMF8	IMF9	IMF10
KELM	0.3634	0.3752	0.2082	0.1318	0.3282	0.1552	0.3303	0.1341	0.1601	0.3465	0.0033
SVR	0.0068	0.0033	0.0038	0.0032	0.003	0.0031	0.0025	0.0023	0.0023	0.0023	0.0027
LSTM	0.0069	0.0034	0.0039	0.0032	0.0029	0.0028	0.0024	0.0024	0.0023	0.0023	0.0028
Trans-former	0.0091	0.0037	0.0039	0.0025	0.0026	0.0017	0.0008	0.0006	0.0003	**0.0002**	0.0027
Informer	**0.0063**	**0.0025**	**0.0027**	0.0032	0.0029	0.0027	0.0024	0.0023	0.0023	0.0022	0.003
GRU	0.0144	0.0055	0.0029	**0.0015**	**0.0006**	**0.0010**	**0.0005**	**0.0002**	**0.0002**	**0.0002**	**0.0001**

with the final prediction results obtained through linear summation. The prediction results for SZ1 are presented in Fig. 7. As illustrated in Fig. 7, the proposed prediction method shows overall excellent performance in forecasting the returns of SZ1. By leveraging the strong predictive capabilities of the decomposed IMF sequences, this approach enables timely and accurate predictions of both trends and numerical changes in actual values, resulting in a high correlation between the real and predicted SZ1 returns. Figure 7 further illustrates the prediction errors, revealing that over 85% of absolute errors are concentrated within 0.01, which underscores the method's robust adaptability to rapid fluctuations in return values Table 3.

4.5 Comparison of Models

To demonstrate the predictive accuracy of the proposed model, this study compares 28 models based on RMSE, MAE and MCS.

(a) To validate the superiority of the CEEMDAN decomposition algorithm, this paper compares it with three other decomposition methods: EEMD, VMD, and SSA. Table 4 presents the evaluation metrics under different decomposition algorithms, while Fig. 8 illustrates the error metrics for these methods. For instance,

Table 4 Evaluation metrics for each model under the SZ1 dataset

	SZ1						
Model	RMSE	MAE	MCS	Model	RMSE	MAE	MCS
CEEMDAN-KLEM	2.2253	2.4915	0.0054	VMD-KELM	2.0684	2.2820	0.0045
CEEMDAN-SVR	0.0076	0.0105	0.4230	VMD-SVR	0.0077	0.0106	0.4325
CEEMDAN-LSTM	0.0073	0.0104	0.4201	VMD-LSTM	0.0076	0.0107	0.4118
CEEMDAN-Transformer	0.0031	0.0023	0.8302	VMD-Transformer	0.0065	0.0049	0.6145
CEEMDAN-GRU	0.0060	0.0094	0.5601	VMD-GRU	0.0062	0.0092	0.6025
CEEMDAN-Informer	0.0026	0.0019	0.8904	VMD-Informer	0.0060	0.0094	0.6574
CEEMDAN-GRU/ Informer	**0.0007**	**0.0005**	**1.0000**	VMD-GRU/ Informer	0.0031	0.0023	1.0000
EEMD-KELM	2.8312	3.1966	0.0042	SSA-KELM	2.5150	2.8706	0.0032
EEMD-SVR	0.0077	0.0106	0.5201	SSA-SVR	0.0112	0.0185	0.4036
EEMD-LSTM	0.0098	0.0145	0.4635	SSA-LSTM	0.0099	0.0150	0.5809
EEMD-Transformer	0.0225	0.0164	0.3548	SSA-Transformer	0.0113	0.0082	0.5723
EEMD-GRU	0.0101	0.0161	0.2635	SSA-GRU	0.0078	0.0107	0.5567
EEMD-Informer	0.0175	0.0131	0.2317	SSA-Informer	0.0031	0.0023	0.9036
EEMD-GRU/Informer	0.0099	0.0150	1.0000	SSA-GRU/Informer	0.0026	0.0019	1.0000

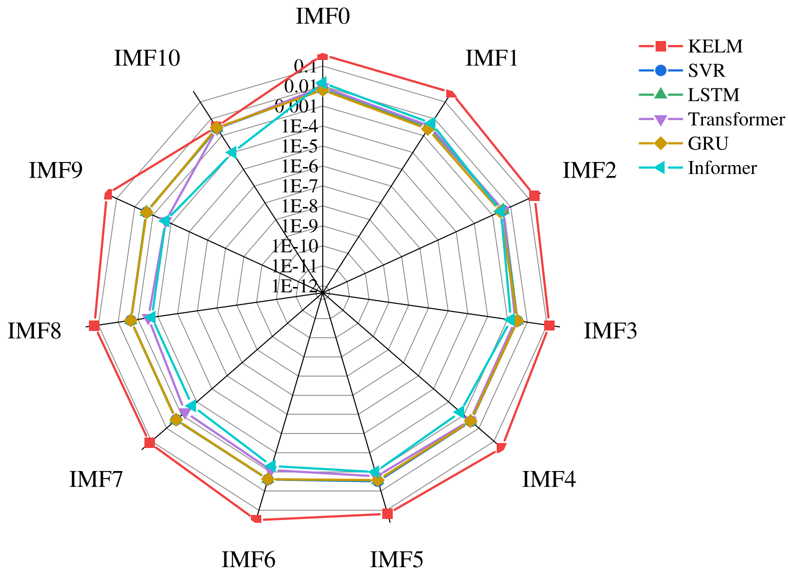


Fig. 8 lg(RMSE) of different prediction models at different components under SZ1 dataset

using SVR as the predictive method on the SZ1 dataset, the MAE of CEEMDAN-SVR is reduced by 7.4%, 8.2%, and 43.02% compared to VMD-SVR, EEMD-SVR, and SSA-SVR, respectively. Similarly, when employing GRU-Informer as the predictive method, the RMSE of CEEMDAN-GRU-Informer decreases by 78.22%, 96.67%, and 73.02% compared to VMD-GRU-Informer, EEMD-GRU-Informer, and SSA-GRU-Informer. The results indicate that CEEMDAN performs the best among the four decomposition algorithms, primarily due to

Table 5 Thresholds for multi-data

	SZ1	SZ6	SH	HSI	DJI	SPX
Threshold	3.2788	1.7283	2.8803	1.3349	1.5012	1.7283

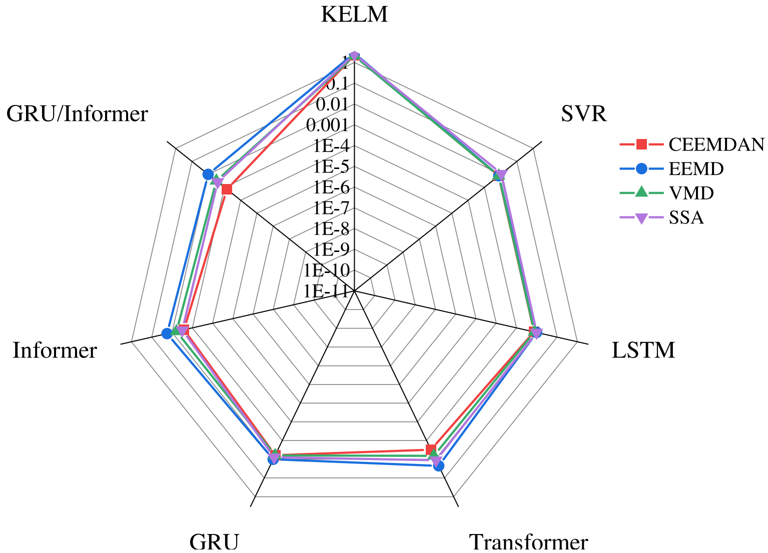


Fig. 9 lg(RMSE) of different predictive models with different decompositions under SZ1 dataset

its improved noise handling mechanism. Unlike the additive noise approach of EEMD, CEEMDAN effectively addresses the modal aliasing problem. Furthermore, CEEMDAN demonstrates strong adaptability, allowing it to select parameters based on varying data characteristics, making it more applicable than VMD and SSA Table 5.

(b) To evaluate the predictive performance of the GRU and Informer algorithms across different frequency components, this paper compares them with four classical predictive models: LSTM, Transformer, KELM, and SVR. Table 3 shows the RMSE values for each component across different predictive models, while Fig. 9 presents the error metrics for these methods. From Table 6, it is evident that GRU has the lowest RMSE for IMF0-IMF2. Taking IMF1 as an example, GRU's RMSE is reduced by 99.33%, 24.24%, 26.47%, 32.43%, and 54.55% compared to KELM, SVR, LSTM, Transformer, and Informer, respectively. In the prediction of IMF8, Informer's RMSE decreases by 99.85%, 80%, 79.17%, 37.5%, and 79.17% compared to KELM, SVR, LSTM, Transformer, and GRU. Experimental results reveal that GRU excels in predicting low-frequency components, while Informer is more suitable for forecasting high-frequency components. This is attributed to GRU's gating mechanism, which effectively captures both short-term and long-term relationships in time series data. Compared to LSTM, GRU has a simpler structure that helps avoid overfitting. The Informer model incorporates a self-attention mechanism, allowing it to focus on multiple

Table 6 Evaluation metrics for each model under the SZ1 dataset

	DJI			SZ6			SPX		
Model	RMSE	MAE	MCS	RMSE	MAE	MCS	RMSE	MAE	MCS
CEEMDAN-KLEM	1.2278	1.3682	0.0803	1.5297	1.6967	0.0624	1.5084	1.6907	0.0654
CEEMDAN-SVR	0.0048	0.0077	0.7856	0.0095	0.0133	0.5897	0.0046	0.0074	0.7889
CEEMDAN-LSTM	0.0035	0.0067	0.8857	0.0082	0.0121	0.6025	0.0044	0.0076	0.8794
CEEMDAN-Transformer	0.0083	0.0052	0.5463	0.0091	0.0067	0.5467	0.0050	0.0032	0.7845
CEEMDAN-GRU	0.0034	0.0066	0.8956	0.0065	0.0123	0.5974	0.0035	0.0063	0.9036
CEEMDAN-Informer	0.0047	0.0078	0.8541	0.0020	0.0013	0.9521	0.0050	0.0033	0.8206
CEEMDAN-GRU/ Informer	**0.0009**	**0.0005**	**1.0000**	**0.0011**	**0.0009**	**1.0000**	**0.0006**	**0.0004**	**1.0000**
EEMD-KELM	1.7510	1.9935	0.0785	1.7405	1.9720	0.0641	0.2665	0.2994	0.1567
EEMD-SVR	0.0048	0.0078	0.7857	0.0096	0.0134	0.3214	0.0047	0.0074	0.8674
EEMD-LSTM	0.0061	0.0102	0.5874	0.0105	0.0148	0.2879	0.0047	0.0074	0.8675
EEMD-Transformer	0.0167	0.0105	0.2541	0.0284	0.0207	0.1355	0.0157	0.0100	0.2564
EEMD-GRU	0.0066	0.0118	0.6574	0.0143	0.0243	0.1256	0.0048	0.0075	0.8612
EEMD-Informer	0.0231	0.0181	0.1564	0.0287	0.0205	0.1356	0.0243	0.0200	0.1895
EEMD-GRU/Informer	0.0035	0.0067	0.8874	0.0224	0.0171	0.2578	0.0040	0.0070	0.8745
VMD-KELM	1.6815	1.8981	0.1054	1.7411	1.8625	0.0874	1.8185	2.0461	0.0751
VMD-SVR	0.0049	0.0077	0.8621	0.0094	0.0131	0.5312	0.0046	0.0073	0.8531
VMD-LSTM	0.0047	0.0078	0.8629	0.0086	0.0127	0.5234	0.0046	0.0075	0.8654
VMD-Transformer	0.0053	0.0034	0.8401	0.0208	0.0172	0.1021	0.0079	0.0050	0.5647
VMD-GRU	0.0041	0.0073	0.8954	0.0076	0.0114	0.5698	0.0040	0.0070	0.7845
VMD-Informer	0.0053	0.0035	0.8861	0.0141	0.0103	0.1543	0.0081	0.0053	0.5589
VMD-GRU/Informer	0.0033	0.0018	0.8994	0.0011	0.0009	0.9857	0.0035	0.0063	0.8956
SSA-KELM	0.1649	0.1846	0.2002	0.1507	0.1723	0.0925	2.4843	2.8117	0.0562
SSA-SVR	0.0048	0.0079	0.8745	0.0095	0.0134	0.5634	0.0047	0.0075	0.8597
SSA-LSTM	0.0049	0.0078	0.8687	0.0096	0.0134	0.5478	0.0051	0.0085	0.8401
SSA-Transformer	0.0084	0.0053	0.5641	0.0141	0.0102	0.2103	0.0079	0.0050	0.6541
SSA-GRU	0.0049	0.0078	0.8689	0.0096	0.0133	0.5417	0.0066	0.0120	0.7452
SSA-Informer	0.0084	0.0053	0.5642	0.0187	0.0149	0.2316	0.0079	0.0050	0.6548
SSA-GRU/Informer	0.0034	0.0066	0.8957	0.0065	0.0123	0.7524	0.0018	0.0013	0.9703

types of information simultaneously, making it particularly adept at capturing short-term fluctuations and dynamic features in high-frequency components.

(c) To illustrate the superior performance of the GRU-Informer architecture, this paper compares the predictive performance of 28 models, as shown in Table 4. The table indicates that GRU-Informer performs exceptionally well. For example, CEEMDAN-GRU-Informer shows RMSE reductions of 94.70%, 73.02%, 78.22%, and 95.18% compared to CEEMDAN-GRU, CEEMDAN-Informer, CEEMDAN-Transformer, and CEEMDAN-LSTM, respectively. The experiments demonstrate that the joint prediction of GRU-Informer exhibits significantly more robust performance than individual models. This is due to the combined strengths of GRU in capturing long-term features and Informer in addressing short-term fluctuations, resulting in superior predictive capabilities.

4.6 Additional Case

To further validate the applicability of our model, we conducted additional experiments in this study using the SZ6, SPX, HSI, N225, SH, and DJI datasets. We collected actual stock price data from the same time frame and applied the same model used in the previous experiments. The results for point predictions and interval predictions are detailed in Tables 6 and 7, respectively. Figure 10 illustrates the prediction outcomes of the models proposed in this paper across the seven datasets. The findings from this additional experiment yield conclusions consistent with those observed for the SZ1 dataset. For the SZ6 dataset, our model achieved a Root Mean Square Error (RMSE) of 0.0011 and a Mean Absolute Error (MAE) of 0.009. In the case of the SPX dataset, the CEEMDAN-GRU-Informer model achieved RMSE and MAE values of 0.0006 and 0.0004, respectively. Compared to the CEEMDAN-GRU, CEEMDAN-Informer, CEEMDAN-Transformer, and CEEMDAN-LSTM models, our model reduced the RMSE by 88%, 82.70%, 88%, and 86.48%, respectively.

Table 7 Multi-method prediction error for SH,N225 and HSI

	SH			N225			HS1		
Model	RMSE	MAE	MCS	RMSE	MAE	MCS	RMSE	MAE	MCS
CEEMDAN-KLEM	2.3612	2.6246	0.0512	0.9229	1.1003	0.1002	0.6135	0.7326	0.1564
CEEMDAN-SVR	0.0053	0.0077	0.8421	0.0134	0.0191	0.2256	0.0068	0.0094	0.7542
CEEMDAN-LSTM	0.0049	0.0070	0.8874	0.0129	0.0193	0.2305	0.0052	0.0077	0.8623
CEEMDAN-Transformer	0.0082	0.0058	0.5478	0.0195	0.0136	0.2564	0.0056	0.0083	0.8426
CEEMDAN-GRU	0.0045	0.0069	0.8898	0.0082	0.0152	0.6580	0.0047	0.0078	0.8932
CEEMDAN-Informer	0.0091	0.0063	0.5124	0.0232	0.0178	0.9862	0.0053	0.0084	0.8564
CEEMDAN-GRU/Informer	**0.0010**	**0.0008**	**1.0000**	**0.0013**	**0.0010**	**1.0000**	**0.0007**	**0.0005**	**1.0000**
EEMD-KELM	3.3654	3.7493	0.0415	0.4202	0.5121	0.1894	2.1833	2.4964	0.2641
EEMD-SVR	0.0055	0.0078	0.8102	0.0136	0.0194	0.2849	0.0068	0.0094	0.7426
EEMD-LSTM	0.0075	0.0105	0.6621	0.0138	0.0194	0.2785	0.0077	0.0112	0.6584
EEMD-Transformer	0.0165	0.0112	0.2574	0.0388	0.0268	0.1526	0.0203	0.0152	0.2897
EEMD-GRU	0.0077	0.0130	0.6619	0.0136	0.0194	0.2879	0.0084	0.0144	0.6845
EEMD-Informer	0.0201	0.0148	0.1972	0.0388	0.0267	0.1257	0.0203	0.0152	0.2954
EEMD-GRU/Informer	0.0045	0.0069	0.8956	0.0100	0.0160	0.2462	0.0052	0.0077	0.8675
VMD-KELM	0.8205	0.9045	0.1035	0.8909	1.0448	0.1021	0.7067	0.8128	0.1987
VMD-SVR	0.0054	0.0078	0.8241	0.0132	0.0190	0.2358	0.0068	0.0093	0.7865
VMD-LSTM	0.0054	0.0077	0.8239	0.0112	0.0167	0.2452	0.0056	0.0083	0.8469
VMD-Transformer	0.0050	0.0050	0.8547	0.0120	0.0088	0.6325	0.0060	0.0044	0.7983
VMD-GRU	0.0054	0.0079	0.8240	0.0100	0.0160	0.2461	0.0053	0.0084	0.8536
VMD-Informer	0.0059	0.0043	0.8213	0.0140	0.0106	0.2856	0.0060	0.0152	0.7954
VMD-GRU/Informer	0.0045	0.0069	0.8957	0.0026	0.0019	0.9624	0.0026	0.0019	0.9567
SSA-KELM	3.7695	4.0066	0.0416	2.2551	2.5799	0.0502	0.2710	0.3176	0.3658
SSA-SVR	0.0053	0.0078	0.8021	0.0134	0.0192	0.1564	0.0069	0.0095	0.7860
SSA-LSTM	0.0054	0.0078	0.8013	0.0146	0.0225	0.2698	0.0069	0.0095	0.7862
SSA-Transformer	0.0082	0.0055	0.6841	0.0194	0.0135	0.2547	0.0099	0.0072	0.5243
SSA-GRU	0.0045	0.0069	0.8952	0.0157	0.0262	0.2569	0.0070	0.0095	0.7845
SSA-Informer	0.0093	0.0065	0.5324	0.0210	0.0152	0.1657	0.0115	0.0091	0.2598
SSA-GRU/Informer	0.0037	0.0027	0.9523	0.0082	0.0152	0.6845	0.0047	0.0078	1.0000

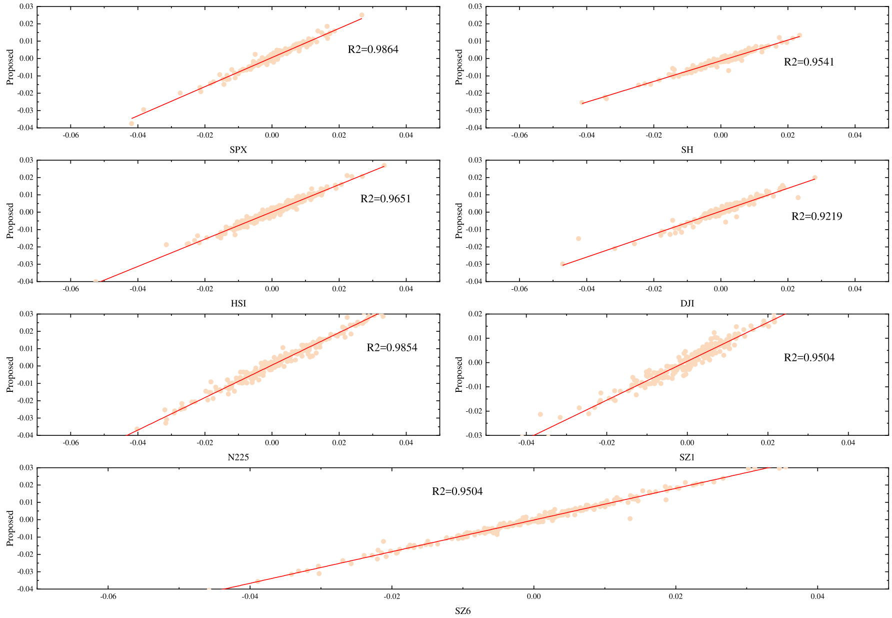


Fig. 10 Prediction results of this paper's model under seven datasets

Consequently, our model demonstrates the highest prediction accuracy among all the benchmark models tested.

5 Conclusion

Stock price forecasting plays a pivotal role in modern financial markets, particularly in light of the rapid evolution of global capital markets and the growing number of market participants. Accurate predictions are vital not only for investors but also for enterprises, influencing investment strategies, risk management, and financing decisions. This study presents a CEEMDAN-GRU-Informer hybrid model, effectively integrating advanced decomposition techniques with deep learning frameworks to improve forecasting accuracy and robustness. The following key findings emerge from the application of the model to six major stock indices across three developed markets:

1. CEEMDAN Decomposition: The CEEMDAN (Complete Ensemble Empirical Mode Decomposition with Adaptive Noise) algorithm proves advantageous in overcoming challenges such as parameter selection and modal effects encountered in other decomposition methods like VMD, SSA, and EEMD. By better isolating relevant components of financial time series, CEEMDAN enhances the model's ability to extract meaningful patterns, ultimately leading to more accurate price predictions.

2. Model Performance: Comparative analyses show that the Gated Recurrent Unit (GRU) model is particularly effective at capturing long-term trends and low-frequency components in financial data. Conversely, the Informer model excels at predicting short-term movements and high-frequency components, showcasing its strength in tracking rapid fluctuations in stock prices. Together, these models provide a more comprehensive approach to forecasting across varying market conditions.
3. Hybrid Model Effectiveness: The combined GRU-Informer approach demonstrates robust performance across multiple datasets, overcoming the common limitations of single-model approaches, such as poor adaptability and overfitting. This hybrid model offers improved generalization and provides more reliable forecasts for investors and market analysts alike.

Based on these findings, this study offers several practical insights for real-world investors:

(1) Long-term Investment Strategy: For investors with a focus on long-term growth, the GRU model's ability to predict low-frequency components can be leveraged to identify underlying trends in stock prices, such as economic cycles, sectoral shifts, or company performance. By focusing on these long-term signals, investors can make strategic investment decisions that align with market fundamentals.
(2) Short-term Trading Decisions: On the other hand, for short-term traders and those seeking to capitalize on price fluctuations, the Informer model's proficiency in forecasting high-frequency components can provide valuable insights into rapid market movements. These forecasts can aid in making timely trading decisions, such as when to enter or exit a stock position based on imminent market shifts.
(3) Risk Management and Diversification: The hybrid model's robustness in handling both long-term trends and short-term volatility is particularly useful in managing risk and optimizing portfolio diversification. Investors can use the GRU-Informer approach to balance their portfolios between more stable, long-term holdings and shorter-term, more volatile investments. This allows for a more nuanced risk management strategy, reducing the impact of market volatility while still capturing profitable opportunities.
(4) Market Timing and Decision Support: The ability to combine both long-term and short-term predictions offers a more accurate picture of the market, enabling investors to time their entry and exit points more effectively. By relying on the hybrid model's forecasts, investors can better align their actions with market cycles, improving the potential for higher returns.

In future research, further exploration of decomposition algorithms and model enhancements will continue to refine our understanding of financial markets, providing even more powerful tools for investors to navigate the complexities of global stock exchanges. With these improvements, investors can make more informed, data-driven decisions, ultimately leading to better investment outcomes.

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